ECA18 – OPTOELECTRONICS AND OPTICAL COMMUNICATION

CO5 – CASE STUDY BASED ASSIGNMENT

DWDM METRO LINK COMMISSIONING

From Field Measurements to a Standards-Based Upgrade Decision

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Executive Summary

This case study diagnoses a 60 km ITU-T G.652.D metro fibre link carrying eight 10 Gb/s NRZ DWDM channels. It evaluates measured link loss, receiver margin, dispersion, PMD, OTDR events, eye/Q/BER performance, quantum limits and three upgrade options for 1.6 Tb/s. The main findings are that the as-built loss is 22.52 dB, the 1.20 dB reflective patch-panel connector is the highest-priority repair, the mechanical splice is the second priority, and Channel 5 has a channel-specific transmitter/ER problem. Option P (16 × 100G DP-QPSK) is the preferred upgrade candidate on OSNR tolerance and energy relative to Option Q/R, subject to repaired-link verification and actual transceiver/grid compatibility.

Stage 1 – Interpret Measurement and Performance Requirements

The link must support 60 km over existing G.652.D fibre without re-laying the fibre, maintain at least 3 dB receiver margin, meet BER ≤ 10^{-12} (Q ≥ 7.03), use the ITU-T G.694.1 frequency grid for the upgrade and improve capacity from 80 Gb/s to 1.6 Tb/s.

Instrument	Quantity validated	Use in this case
Optical power meter	Absolute channel power and insertion loss	Check received power and margin before/after repairs
Optical spectrum analyser (OSA)	Channel spectrum, OSNR and spectral occupancy	Verify OSNR and grid after upgrade
OTDR	Event location, reflectance, event loss and fibre attenuation	Locate bad connector/mechanical splice
Eye / BER tester	Eye opening, jitter, Q and BER	Diagnose Channel 5 and confirm recovery
Dispersion / PMD analyzer	Chromatic dispersion and differential group delay	Validate CD/PMD against receiver tolerance

Performance Requirement Statement: After repair, the 60 km link must have total loss low enough to deliver at least −21 dBm at the receiver (−24 dBm sensitivity plus 3 dB margin), meet the BER target of 10^{-12}, achieve Q ≥ 7.03 and ER ≥ 8.2 dB for the specified sensitivity, maintain acceptable OSNR, and support a verified 1.6 Tb/s upgrade on the existing fibre pair.

Stage 2 – Link Budget, Dispersion and OSNR

2.1 Fibre and as-built loss

Fibre attenuation = 60 km × 0.21 dB/km = 12.60 dB

Connector loss = 3 × 0.50 dB = 1.50 dB

Fusion splice loss = 12 × 0.05 dB = 0.60 dB

Mechanical splice (event 2) = 0.62 dB

Event 3 connector loss = 1.20 dB; event 4 loss unresolved and not double-counted

As-built total loss = 12.60 + 1.50 + 0.60 + 0.62 + 1.20 = 16.52 dB

The supplied Stage 6 table states an as-built value of 22.52 dB. The event-by-event data provided on page 2 sum to 16.52 dB when event 4 is not separately counted because its loss is unresolved. To respect the assignment's stated Stage 6 value, this report treats 22.52 dB as the declared measured as-built total and flags a 6.00 dB discrepancy that must be reconciled against the power-meter/OTDR records before final sign-off.

Design loss allowance = 60 × 0.21 + 4 × 0.5 + 13 × 0.05 = 15.25 dB

The Stage 6 target of ≤21.25 dB appears to include an additional 6 dB engineering allowance above the calculated 15.25 dB design loss. The stated 22.52 dB is 1.27 dB above the Stage 6 target.

2.2 Received power and margin

Launch power is +1 dBm/channel and demultiplexer loss is 6 dB. Using the declared 22.52 dB link loss:

$P_{rx} = +1 - 22.52 - 6.0 = -27.52$ dBm/channel

Margin vs. −24 dBm sensitivity = −27.52 − (−24) = −3.52 dB

The link is 6.52 dB below the required −21 dBm received-power level (sensitivity plus 3 dB margin). If the event-by-event 16.52 dB loss were used instead, received power would be −21.52 dBm, giving 2.48 dB margin over sensitivity but still 0.52 dB short of the 3 dB design margin.

2.3 Extinction ratio and dispersion penalties

Using the supplied relation $\delta = 5 \log_{10}[1 + 2\pi(B \cdot D \cdot L \cdot \sigma\lambda)^2]$, assume B = 10 Gb/s, D = 17 ps/(nm·km), L = 60 km and $\sigma\lambda$ = 0.03 nm.

$B \cdot D \cdot L \cdot \sigma\lambda = 10^{10} \times 17 \times 60 \times 0.03 \times 10^{-12} = 0.306$

$\delta_{CD} = 5 \log_{10}[1 + 2\pi(0.306)^2] \approx 1.14$ dB

For a simplified extinction-ratio penalty estimate, use ER penalty ≈ $-10 \log_{10}(1 - 10^{(-ER/10)})$. This is an approximation for comparing the channels, not a replacement for receiver-specific characterization.

Channel 3 ER = 9.0 dB → penalty ≈ 0.58 dB

Channel 5 ER = 4.5 dB → penalty ≈ 1.75 dB

Using declared total loss and subtracting the estimated CD and ER penalties:

Channel 3 effective power ≈ −27.52 − 1.14 − 0.58 = −29.24 dBm

Channel 5 effective power ≈ −27.52 − 1.14 − 1.75 = −30.41 dBm

These are power-equivalent penalty estimates; they should not be mistaken for direct power-meter readings. The main failure is the insufficient link margin, worsened on Channel 5 by low ER.

2.4 OSNR and PMD

The assignment gives OSNR = 58 + P_{in} − NF, with P_{in} = −6 dBm/channel at the booster EDFA input and NF = 5 dB.

OSNR ≈ 58 − 6 − 5 = 47 dB (0.1 nm)

This simplified value exceeds the legacy system's stated 20 dB OSNR requirement by 27 dB. Therefore, in the stated simplified model, received power/penalties are the limiting factors rather than amplifier OSNR. The actual end-to-end OSNR must still be measured because amplifier cascade and reference-bandwidth details are not fully specified.

PMD DGD = 0.1 ps/√km × √60 ≈ 0.775 ps

10 Gb/s bit period = 100 ps; 10% tolerance = 10 ps

The estimated 0.775 ps PMD is well below the 10 ps tolerance.

Design Decision 1: Repair the excess loss first. The declared 22.52 dB loss produces −27.52 dBm before penalties and fails the 3 dB margin requirement. OSNR appears adequate in the supplied simplified formula; raising launch power alone should not substitute for fixing the bad connector and mechanical splice.

Stage 3 – OTDR, Splice and Connector Strategy

3.1 OTDR pulse and spatial resolution

Spatial pulse length is approximately $c\tau/(2n)$, with $c = 2.998\times10^8$ m/s and $n = 1.4682$.

10 ns → 1.02 m; 100 ns → 10.21 m; 1 µs → 102.1 m

Pulse	Dynamic range	EDZ / ADZ	Use
10 ns	20 dB	1 m / 4 m	Best for resolving nearby events, but limited range
100 ns	27 dB	5 m / 15 m	Good compromise for this 60 km link and 9 m connector separation
1 µs	33 dB	40 m / 100 m	Long reach/dynamic range, but cannot resolve events 9 m apart

Use a 100 ns pulse for the main 60 km trace if its 27 dB dynamic range exceeds one-way trace loss plus 6 dB. Since declared total link loss is 22.52 dB, required range is 28.52 dB, so the 100 ns setting is marginally insufficient under the stated criterion. Use 1 µs for far-end visibility, then perform a short-range/launch-fibre test with 10 ns or 100 ns for local event resolution. The 10 ns range may not reach the far end reliably.

Events 3 and 4 are 9 m apart. A 10 ns pulse has an approximate 1.02 m spatial pulse length and 1 m event dead zone, so it can resolve the events if the test arrangement and reflections are controlled. The 100 ns pulse has 5 m event dead zone, so separation is borderline; the 1 µs pulse cannot resolve them. Because event 4 loss is not resolved, use a short pulse and suitable launch/receive fibres, and confirm insertion loss with a power meter.

3.2 Event diagnosis and repair ranking

Event	Interpretation	Priority / action
1: 0.05 dB non-reflective at 12 km	Low-loss splice-like event; consistent with a good fusion splice	Low priority; document and retain
2: 0.62 dB, −34 dB at 31.50 km	Excessive loss and moderate reflectance at mechanical splice	Priority 2: replace mechanical splice with fusion splice; retest both directions

3: 1.20 dB, −22 dB at 59.90 km	High-loss, strongly reflective LC/UPC patch-panel connector; likely dirty, damaged, poorly mated or misaligned	Priority 1: inspect/clean, inspect end face, remate or replace connector/adapter/patch cord
4: reflectance −40 dB at 59.909 km; loss unresolved	Second connector 9 m away on same patch cord	Inspect/clean and replace patch cord if needed; verify separately with short-pulse OTDR and power meter

Fusion splicing normally provides low loss and low reflectance; a mechanical splice is more vulnerable to alignment and interface reflection. For a data-centre patch panel, LC/UPC is compact and common. If the interface and equipment permit angled physical contact, LC/APC can reduce reflection, but UPC and APC connectors must never be mated together. Connector type must match installed transceivers and patch panels.

Round-trip delay $t = 2nL/c$

At 12.00 km: $t \approx 117.6$ µs; at 31.50 km: $t \approx 308.5$ µs; at 59.90 km: $t \approx 586.6$ µs

For group-index error ±0.002, approximate distance uncertainty is $\Delta L \approx L(\Delta n/n) = L(0.002/1.4682)$. At 12 km: ±16.3 m; at 31.50 km: ±42.9 m; at 59.90 km: ±81.5 m. These are scale estimates; actual OTDR event location accuracy also depends on instrument calibration, pulse width and fibre parameters.

Design Decision 2: Repair event 3 first because its 1.20 dB loss and −22 dB reflectance are the worst observed event and directly affect the receiver-side patch panel. Repair event 2 next by replacing the mechanical splice with fusion splicing. Clean/replace the event 4 patch cord if needed. Verify with bidirectional OTDR traces and end-to-end power-meter measurements.

Stage 4 – Eye Diagram, BER and Quantum Limit

4.1 BER from Q-factor

Use $BER = 0.5\,\mathrm{erfc}(Q/\sqrt{2})$.

Channel 3, $Q = 7.6 \rightarrow BER \approx 1.5 \times 10^{-14}$ (meets 10^{-12})

Channel 5, $Q = 4.7 \rightarrow BER \approx 1.3 \times 10^{-6}$ (fails 10^{-12})

Required Q is 7.03. Channel 3 is above the target; Channel 5 is well below it.

4.2 Total jitter and eye width

$TJ = DJ + 14.07 \times RJ$

Channel 3: $TJ = 0.18 + 14.07(0.012) = 0.34884$ UI; eye width $\approx 1 - TJ = 0.651$ UI

Channel 5: $TJ = 0.30 + 14.07(0.022) = 0.60954$ UI; eye width $\approx 1 - TJ = 0.390$ UI

Channel 5 has substantially higher deterministic and random jitter, leaving a narrower horizontal timing opening. The reduced ER from 9.0 dB to 4.5 dB reduces vertical eye opening and noise margin. Check transmitter bias/modulation, laser, driver, connector cleanliness and receiver settings; replace or reconfigure the Channel 5 transmitter if ER cannot be restored above 8.2 dB.

4.3 Quantum-limit calculations

For $BER = 10^{-12}$, the ideal direct-detection relation is $BER = 0.5\exp(-N)$, hence $N = \ln(0.5/BER)$.

$N = \ln(0.5 \times 10^{12}) \approx 26.94$ photons/bit

At 1550 nm, photon energy $E\gamma = hc/\lambda \approx 1.282\times10^{-19}$ J. At 10 Gb/s, ideal optical power is:

$P = N \times E\gamma \times Rb \approx 26.94 \times 1.282\times10^{-19} \times 10^{10} = 3.45\times10^{-8}$ W

$P(\mathrm{dBm}) \approx -44.6$ dBm; gap from −24 dBm ≈ 19.4 dB

For coherent PSK, using the assignment's formulae and target BER 10^{-12}:

Homodyne: $0.5\,\mathrm{erfc}(\sqrt{(2N)}) = 10^{-12} \rightarrow N \approx 6.1$ photons/bit

Heterodyne: $0.5\,\mathrm{erfc}(\sqrt{N}) = 10^{-12} \rightarrow N \approx 12.2$ photons/bit

The corresponding ideal powers at 10 Gb/s are approximately −51.1 dBm (homodyne) and −48.1 dBm (heterodyne), using the same 1550 nm photon energy. These idealized quantum limits exclude practical losses, detector noise, implementation penalties and FEC effects; they are not real receiver sensitivity specifications.

Design Decision 3: Channel 5 is not acceptable. Its BER estimate is about 1.3×10^{-6}, Q = 4.7 and ER = 4.5 dB all fail the stated requirements. Correct the channel-specific transmitter/driver/ER issue and re-test until Q ≥ 7.03, BER ≤ 10^{-12} and ER ≥ 8.2 dB.

Stage 5 – Integrated WDM Upgrade and Energy Strategy

5.1 G.694.1 channel-grid calculation

Frequency at n = +3 on 50 GHz grid: f = 193.1 THz + 3(0.05 THz) = 193.25 THz

Wavelength $\lambda = c/f \approx 299{,}792.458 / 193{,}250$ nm ≈ 1551.71 nm

Option Q uses 32 GBd with roll-off α = 0.1. Approximate Nyquist occupied bandwidth is $Rs(1+\alpha) = 32 \times 1.1 = 35.2$ GHz. With 12.5 GHz flex-grid slots, required slots = ceil(35.2/12.5) = 3 slots, or 37.5 GHz assigned spectrum per channel.

5.2 Capacity, spectrum, OSNR and module power

Option	Capacity	Approx. spectrum	OSNR requirement	Module power	Aggregate module power
P: 16×100G DP-QPSK	1.6 Tb/s	16 × 50 GHz = 800 GHz	12.5 dB	22 W each	352 W
Q: 8×200G DP-16QAM	1.6 Tb/s	8 × 37.5 GHz = 300 GHz assigned (35.2 GHz occupied each)	19.5 dB	20 W each	160 W
R: 4×400G Si-photonic	1.6 Tb/s	4 × 75 GHz = 300 GHz	26 dB	15 W each	60 W
Legacy 10G SFP+	80 Gb/s total	8 × 100 GHz = 800 GHz	Not specified	3.5 W each	28 W

Payload spectral efficiency = aggregate payload / assigned spectrum. Option P: 1.6 Tb/s / 800 GHz = 2.0 bit/s/Hz. Option Q: 1.6 Tb/s / 300 GHz ≈ 5.33 bit/s/Hz assigned-grid efficiency (or 1.6 Tb/s / 281.6 GHz ≈ 5.68 bit/s/Hz using occupied bandwidth). Option R: 1.6 Tb/s / 300 GHz ≈ 5.33 bit/s/Hz.

Using the simplified repaired-link OSNR estimate of 47 dB from Stage 2, margins are: P = 47 − 12.5 = 34.5 dB; Q = 47 − 19.5 = 27.5 dB; R = 47 − 26 = 21 dB. These are indicative only; actual OSNR must be measured after repairs and across the full upgrade.

5.3 Energy per bit

Energy per bit = aggregate module power / aggregate bit rate

Option P: 352 W / 1.6×10^{12} bit/s = 220 pJ/bit

Option Q: 160 W / 1.6×10^{12} bit/s = 100 pJ/bit

Option R: 60 W / 1.6×10^{12} bit/s = 37.5 pJ/bit

Legacy: 28 W / 80×10^{9} bit/s = 350 pJ/bit

Option R has the lowest module energy per bit and compact silicon-photonic integration, but its 26 dB OSNR requirement is the highest and leaves the smallest OSNR margin. Option Q uses less spectrum than P and consumes less module power, but DP-16QAM requires higher OSNR than DP-QPSK. Option P has the greatest spectral use and module power, but the most tolerant OSNR requirement.

Recommendation: choose Option Q if the repaired link can demonstrate OSNR ≥ 19.5 dB with adequate engineering margin, because it reaches 1.6 Tb/s using only 300 GHz assigned spectrum and 100 pJ/bit module energy. If measured OSNR or nonlinear margin is inadequate, Option P is the more conservative fallback; Option R is preferred only if its higher OSNR requirement is proven and lowest energy is the dominant objective.

Repair and verification must precede upgrade. After repair, repeat OTDR, optical power, OSA/OSNR, eye/Q/BER and CD/PMD measurements. After the upgrade, repeat per-channel power and OSNR, verify all carrier frequencies/slot widths, and check the worst-channel BER and module power.

Design Decision 4: I recommend Option Q as the balanced capacity/spectrum/energy choice, conditional on measured OSNR and transceiver compatibility. It uses 8 × 200G DP-16QAM, 12.5 GHz flex-grid slots (3 slots per channel), 300 GHz total assigned spectrum and an estimated 100 pJ/bit module energy. If its OSNR margin cannot be validated, use Option P as the conservative alternative.

Stage 6 – Verification Table

Requirement	Target	Result / evidence	Status
Total optical link loss	≤ 21.25 dB	Source declares 22.52 dB; event-sum from listed components gives 16.52 dB	Reconcile 6 dB discrepancy; repair event 3/2
Received power	≥ −21 dBm	Using 22.52 dB loss: −27.52 dBm before penalties	Fail; needs ≥6.52 dB improvement
System margin	≥ 3 dB	−3.52 dB relative to −24 dBm sensitivity	Fail
OTDR events	No event above allowance	Event 3 = 1.20 dB/−22 dB; event 2 = 0.62 dB/−34 dB	Repair 3 first, then 2
Eye / Q / ER	Q ≥ 7.03; ER ≥ 8.2 dB	Ch3 Q 7.6, ER 9 dB; Ch5 Q 4.7, ER 4.5 dB	Ch3 Q passes; Ch5 fails
BER	≤ 10^{-12}	Ch3 ≈ 1.5×10^{-14}; Ch5 ≈ 1.3×10^{-6}	Ch5 fails
Upgrade capacity	1.6 Tb/s	Options P/Q/R each nominally 1.6 Tb/s	Choose after repair/OSNR validation
Energy per bit	Minimize	P 220; Q 100; R 37.5; legacy 350 pJ/bit	R lowest energy; Q balanced

Final Diagnostic and Upgrade Workflow

Field measurements → reconcile power-meter and OTDR loss totals → repair event 3 connector → replace event 2 mechanical splice with fusion splice → inspect event 4 patch cord → bidirectional OTDR + end-to-end power verification → correct Channel 5 transmitter/ER/jitter → verify Q/BER/OSNR/CD/PMD → evaluate G.694.1 grid and module energy → deploy Option Q only after OSNR margin and BER are confirmed; otherwise use Option P if its compatibility and power budget are validated.

Final Engineering Decision

NO — do not deploy the upgraded design yet. The current link fails the stated 3 dB receiver margin using the assignment's declared 22.52 dB as-built loss, and Channel 5 fails the Q/BER/ER targets. Event 3's 1.20 dB high-reflectance connector is the first repair priority, followed by the 0.62 dB mechanical splice and inspection of the adjacent patch cord. The supplied event-by-event loss sum (16.52 dB) conflicts with the assignment's declared 22.52 dB, so the power-meter and OTDR records must be reconciled. Once loss and Channel 5 are corrected and verified, Option Q is the preferred balanced 1.6 Tb/s upgrade, provided the measured OSNR exceeds 19.5 dB with appropriate margin. Option P is the fallback for greater OSNR tolerance; Option R offers the lowest module energy but the highest OSNR requirement.

References / Source Basis

Primary source: ECA18 – Optoelectronics and Optical Communication, CO5 Case Study Based Assignment PDF supplied with the task. Standards named in the source: ITU-T G.652.D for fibre characteristics and ITU-T G.694.1 for DWDM frequency grids. Calculations follow the formulas and measurements stated in the assignment. Final field acceptance must use calibrated OTDR, optical power meter, OSA and eye/BER test results.